Jimmy H. Lynch

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SUMMARY

Physics Ph.D. candidate with industry experience specializing in radio observations of astrophysical transients. Researching the late-time radio emission of tidal disruption events (TDEs) and its connection to fundamental physics of accreting black holes. Interested in utilizing machine learning (ML) to select transients for radio follow-up. Proficient in Python, ML, and radio astronomy imaging (CASA). Experienced research mentor and teacher.

EDUCATION

Ph.D. in Physics — expected 2029
University of Oregon — *Eugene, OR*
Advisor: Prof. Yvette Cendes

M.Sc. in Physics — June 2025
University of Oregon — *Eugene, OR*

B.A. in Physics w. Astrophysics Concentration — May 2021
University of Pennsylvania — *Philadelphia, PA*
magna cum laude + honors thesis

RESEARCH EXPERIENCE

Graduate Research Assistant — Sept 2024 – present
Department of Physics — *University of Oregon*
Advisor: Prof. Yvette Cendes

- Characterizing the late-time radio emission of tidal disruption events (TDEs) from ongoing observing campaigns with the Very Large Array (VLA), the Australian Telescope Compact Array (ATCA), and others
- Developing machine learning frameworks to efficiently prioritize radio follow-up

Research Scientist, Solar Physics (Part-time, 8 hours/week) — July 2023 – Sept 2024
Lockheed Martin Space, Advanced Technology Center — *Palo Alto, CA*
Manager: Dr. Neil Hurlburt, Space Science and Instrumentation Directorate

- Developed a pipeline to efficiently access high-res solar images on the HelioCloud

Research Scientist, Advanced Materials — June 2021 – June 2023
Lockheed Martin Space, Advanced Technology Center — *Palo Alto, CA*
Manager: Dr. Bryan Gockel, Hypersonics and Advanced Materials Directorate

- Served as PI on programs using ML to discover novel metal alloys and supported computational research efforts on laser-material interaction and high-temp materials

Undergraduate Researcher Aug 2020 – May 2021
Department of Physics & Astronomy, University of Pennsylvania *Philadelphia, PA*
Advisor: Prof. James Aguirre

- Researched towards honors thesis titled "A Standardization of Cosmological Simulations 21cmFAST and zreion for Robust Neural Network Training Data Generation"

Research Science Intern May 2020 – Aug 2020
Lockheed Martin, Advanced Technology Center *Palo Alto, CA*
Manager: Dr. Steven Shepard

- Supported research in additive manufacturing and COVID respirator design and testing

Undergraduate Researcher Aug 2019 – Dec 2019
Department of High-Energy Astrophysics, University College Cork *Cork, Ireland*
Advisor: Prof. Paul Callanan

- Classified 16 X-ray sources in cluster Omega Centauri using Hubble and Chandra data

Undergraduate Researcher Jan 2018 – Aug 2019
Department of Mechanical Engineering & Applied Mechanics, University of Pennsylvania *Philadelphia, PA*
Advisors: Prof. Jordan Raney and Dr. Hiromi Yasuda

- Researched the mechanical features of an origami structure for Martian housing units

RESEARCH FUNDING

Awarded $75k for Bright Ideas: Lockheed Martin Corporate Seedling Initiative 2022
Proposal title: "Closed-Loop Artificial Intelligence for Additive Manufacturing" *Palo Alto, CA*

Awarded $100k for Lockheed Martin Internal Research and Development (IRAD) 2022
Proposal title: "Artificial Intelligence for Radiation-Hardened Materials Design" *Palo Alto, CA*

PUBLICATIONS

First Author

1. **J. H. Lynch**, Y. Cendes, N. Velez, C. Christy, K. Alexander, E. Berger, R. Chornock, A. J. Goodwin, T. Laskar, and R. Margutti, "A Decade-Delayed Radio Flare in Tidal Disruption Event ASASSN-14ae", submitted to the *Astrophysical Journal*, 2026.

Major Contributor

- S. P. Shepard, W. W. Yuen, **J. H. Lynch**, J. N. Maser, A. Swaminathan, and D. Fabris, "Experimentation and Modeling of Laser Radiation Scattering Through Carbon Fiber Reinforced Polymers", *AIAA Journal of Heat Transfer and Thermophysics*, May 2022.
- Y. Miyazawa, H. Yasuda, H. Kim, **J. H. Lynch**, K. Tsujikawa, T. Kunimine, J.R. Raney, and J. Yang, "Heterogeneous origami-architected materials with variable stiffness", *Communications Materials*, November 2021.

OBSERVING PROJECTS

Principal Investigator

- **MKT/26037** *Submitted*
  "MeerKAT Observations of Gaia Black Holes with Stellar Companions"
- **VLA/27A-370** *Submitted*
  "VLA Observations of Gaia Black Holes with Massive Star Companions"
- **VLA/27A-352** *Submitted*
  "Searching for Delayed Radio Flares from Tidal Disruption Events"
- **VLA/26B-255** 36 hours
  "A Continued Search for Delayed Radio Emission from Tidal Disruption Events"
- **VLA/26A-370** 23 hours
  "Searching for Secondary Radio Flares from Tidal Disruption Events"
- **VLA/25B-272** 30 hours
  "A Continued Survey for Delayed Radio Emission in Tidal Disruption Events"

Co-Investigator

- **VLA/25B-154** *Submitted*
  "Tracking the long-term radio evolution of a sample of tidal disruption events"
- **VLA/25B-220** *Submitted*
  "Search for Very Late-Time Radio Emission from X-ray Tidal Disruption Events"
- **VLBA/25A-283** 39 hours
  "The outflow speeds and geometries of thermal tidal disruption events"
- **VLA/25A-060** 15 hours
  "Searching for Secondary Flares in Radio Emission from Tidal Disruption Events"
- **VLA/25A-300** 43 hours
  "Prompt Radio Emission in TDEs: Probing the Early Disruption Process"
- **VLA/25A-037** 4.5 hours
  "Continued Observations of a Relativistic Outflow in the TDE AT2018hyz"
- **VLA/24B-249** 20.25 hours
  "A Survey for Delayed Radio Emission in Tidal Disruption Events"
- **VLBA/24B-252** 392 hours
  "The first VLBA Large Survey of Astronomical Transients"
- **VLA/24B-311** 68 hours
  "The first VLBA Large Survey of Astronomical Transients"

AWARDS & RECOGNITION

UO Department of Physics Weiser Graduate Leadership Award	2025
UO Department of Physics Weiser First Year Teaching Award	2024
Advanced Materials Junior Employee of the Quarter	2022
3.8/4.0 Overall TA Rating at Penn (5 semesters, 100+ responses) vs. 2.9/4.0 instructor average	2018 – 2021
Penn Department of Mathematics Good Teaching Award (Fall and Spring)	2020
Recipient of Penn Summer Undergraduate Research Fellowship	2019
1st Place, American Institute of Aeronautics and Astronautics Student Conference	2019
Penn Department of Mathematics Good Teaching Award (Fall)	2018

TEACHING EXPERIENCE

Graduate Teaching Initiative (GTI) Participant Aug 2026 - present
University of Oregon *Eugene, OR*
Through this program, I am developing a teaching portfolio with a teaching philosophy, example curricula, class observations, and course reflections.

Graduate Teaching Assistant, 16 hours/week Sept 2023 – present
Department of Physics, University of Oregon *Eugene, OR*

- ASTR 122: The Solar System (Fall 2025)
- PHYS 253: Electricity and Magnetism (Spring 2024)
- PHYS 202: Intro Physics II (Winter 2024)
- ASTR 121: The Life of Stars (Fall 2023)

Undergraduate Teaching Assistant, 13 hours/week Aug 2018 – May 2021
Department of Mathematics, University of Pennsylvania *Philadelphia, PA*

- Math114: Multivariable Calculus (Fall 2020)
- Math240: Linear Algebra & Differential Equations (Spring 2020, Spring 2021)
- Math104: Calculus I (Fall 2018, Spring 2019)

MENTORSHIP & OUTREACH

High School Research Mentor and Mentor Coordinator Sept 2024 – present
Department of Physics, University of Oregon *Eugene, OR*
I mentor high school students on science projects for their regional science fairs

- **Zoe K.**, awarded 3rd Place and **$36,000 scholarship** to UO, Feb 2025
  Poster: *"Examining Properties of Galaxies Producing Radio TDEs"*
- **Ella H.**, awarded 2nd Place and **$2,000 scholarship** to OSU, March 2025
  Poster: *"The Typicality of the Sun Compared to Other Stars"*

Mentorship Coordinator, Physicists for the Advancement of Gender Equity in Science Apr 2024 – Jan 2026
Department of Physics, University of Oregon *Eugene, OR*
I helped pair undergraduate physics majors with ~20 graduate student mentors based on shared identities with the goal of increasing support for underrepresented groups in STEM.

CONFERENCE PROCEEDINGS

1. **J. H. Lynch**, "Convolutional Autoencoders for Denoising Solar Images", *Python in Science Conference, 2024 (SciPy),* July 2024. DOI: 10.25080/fpju4745
2. **J. H. Lynch** and J. R. Raney, "Volumetric Origami-based Modular Space Structures with Tailorable Stiffness", *2020 SciTech Forum Meeting Papers*, January 2020. DOI: 10.2514/6.2020-0110

CONFERENCE PRESENTATIONS

1. **Tidal Disruption Events and Nuclear Transients in Crete** (Heraklion, Crete, Greece) Aug 2026
   Talk: *"A Decade-Delayed Radio Flare from TDE ASASSN-14ae"*
2. **High Energy Astrophysics Division (HEAD) Conference** (St. Loius, MO) Oct 2025

Poster: *"The Late-Time Radio Brightening of TDE ASASSN-14ae"*

3. **Oregon Astronomy Research Symposium (OARS)** (Eugene, OR) Sept 2025
   Talk: *"The Late-Time Radio Brightening of TDE ASASSN-14ae"*
4. **Solar Heliospheric and INterplanetary Environment Workshop** (Juneau, AK) Aug 2024
   Poster: *"Convolutional Autoencoders for Denoising Solar Images"*
5. **Scientific Computing in Python (SciPy) 2024** (Seattle, WA) July 2024
   Virtual Poster: *"Convolutional Autoencoders for Denoising Solar Images"*
6. **Lockheed Martin Artificial Intelligence Summit** (Virtual) Sept 2022
   Talk: *"Machine Learning for Advanced Materials Discovery"*
7. **Scitech Forum, International Student Conference** (Orlando, FL) Jan 2020
   Talk: *"Origami-based Modular Space Structures with Tailorable Stiffness"*

SKILLS

Programming:	Python, MATLAB, Java, Git, LaTeX, HTML
ML packages:	Tensorflow, Scikit Learn, Torch, zarr, xarray, dask
Physics modeling:	COMSOL Multiphysics, Thermal Desktop, LAMMPS, VASP

GRADUATE COURSEWORK

Theoretical Mechanics	Statistical Mechanics x2	Quantum Mechanics x3
Math Methods of Physics	Electromagnetic Theory x2	General Relativity
Computational Neuroscience	Image Analysis	Cosmology
Intro to Artificial Intelligence	Scientific Computing	